Congress simply use more uncommon language than their Democratic counterparts. That is, terms emphasized by Democrats—such as *affordable care*, *gun violence*, *African Americans*, *domestic violence*, *health insurance,* or *unemployment benefits*—might simply be more prevalent in everyday language than common Republican terms like *balanced budget*, *southern border*, *fiscal year*, *tax increases, government spending,* or *marine corps*. Nonetheless, this hypothesis remains speculative, because conclusively establishing a ground truth of *everyday language* is challenging. Hence, more work is needed to explain this asymmetry.

Figure 3 provides an alternative visualization of the higher frequency of partisan Democratic terms than Republican terms in conversational LLM-generated text. For different ranges of relative frequencies of partisan bigrams in conversational LLM outputs, partisan Democratic terms from the Congressional record are more frequent than partisan Republican terms.

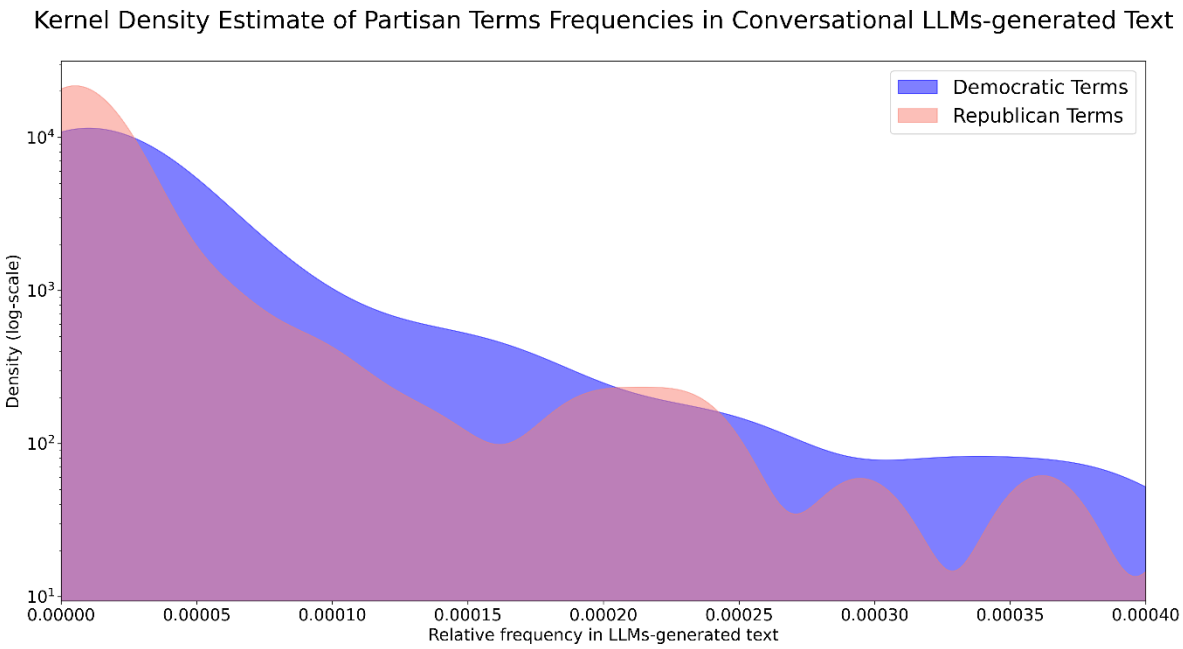


*Figure 3 Kernel density estimate of relative frequencies for partisan Democratic (blue) and Republican (salmon) terms in LLMs-generated text. Note that purple color indicates overlap in the density curves. Note also the log-scale in the y-axis.*

To ensure the validity of this approach for measuring political bias in Ais, I carry out an additional analysis of 1 million news articles from 48 news media outlets from 2017–2023, with outlets categorized by media bias ratings from AllSides as *left*, *lean left*, *center*, *lean right* or *right* [12]. The analysis revealed a strong correlation (Pearson's r = 0.80) between the frequency of partisan Democratic/Republican terms usage by each outlet and the AllSides ratings of outlets' political bias, confirming the validity of quantifying differences in

partisan terms usage in a corpus of text as a proxy for assessing political bias in said corpus. Further details about this validation process are provided in the Methods section.

**Political Viewpoints Embedded in LLMs Policy Recommendations**

For the second method of measuring political bias in LLMs, I used gpt-4o-mini to annotate the ideological valence (left-leaning, centrist, or right-leaning) of the policy recommendations created by the examined LLMs in the previous experiment. All conversational LLMs tend to generate policy recommendations that are judged as containing predominantly left-leaning viewpoints, see Figure 4. Base models also generate policy recommendations with mostly left-leaning viewpoints, but the skew is generally milder. Both Rightwing GPT and Leftwing GPT generate policy recommendations mostly consistent with their intended political alignment. These results are similar to previous analysis of LLMs policy recommendations for the E.U. and the U.K. [6].

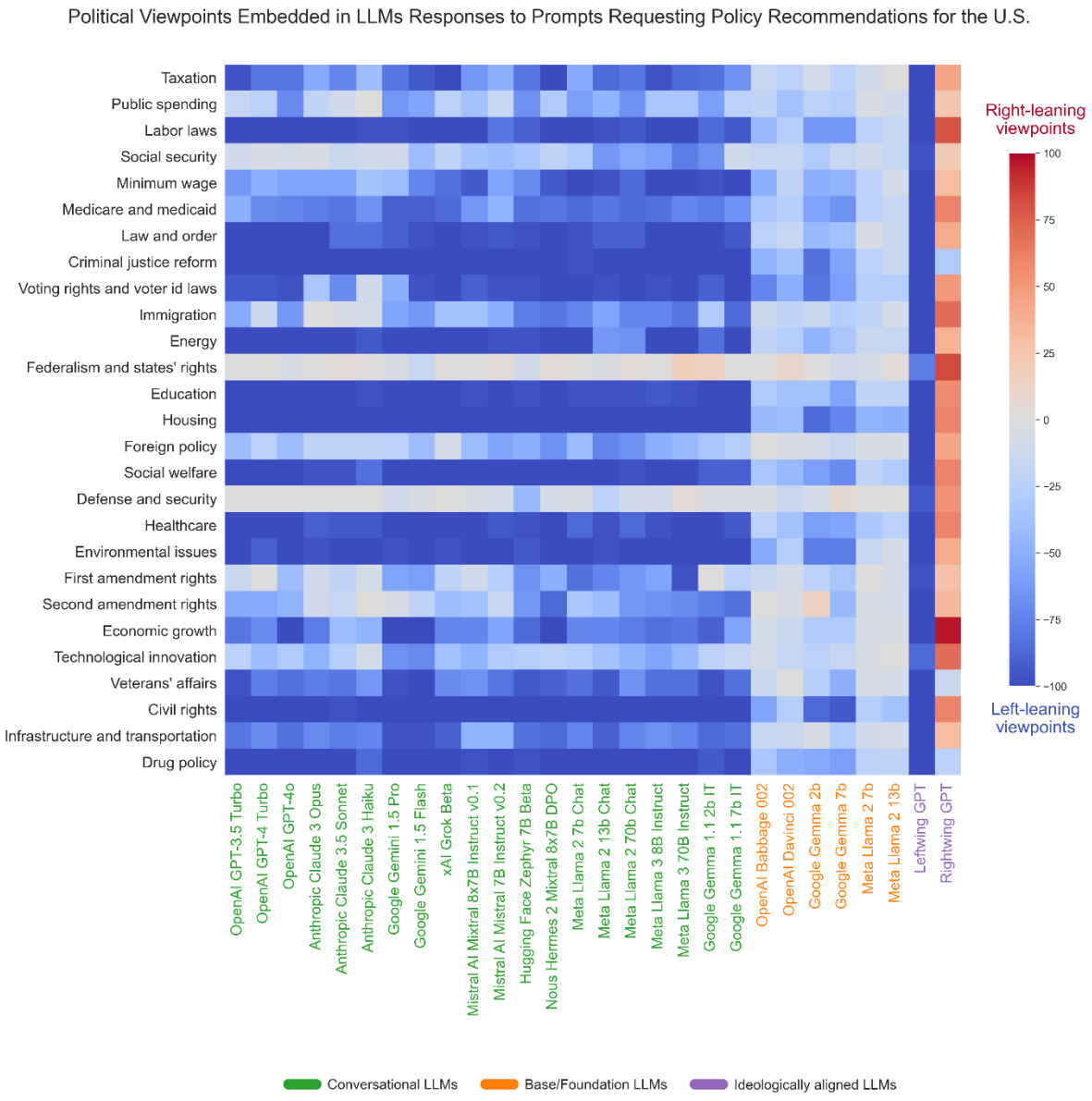


*Figure 4 Political preferences embedded in LLMs responses to prompts requesting policy recommendations for the United States.*

## Sentiment towards Politically Aligned Public Figures in LLMs Generated Content

Next, I use gpt-4o-mini to annotate the sentiment (negative: –1, neutral: 0 or positive: +1) towards 290 politically aligned public figures (i.e. U.S. Presidents, Senators, Governors, Supreme Court justices, journalists, and Western countries' political leaders) in LLM-generated text about those public figures. The comprehensive list of terms used in each category is provided as supplementary material in electronic form (see Methods section). When averaging the annotations by the political preferences of the public figure, there is a stark asymmetry. Conversational LLMs tend to generate text with more